- **Economic Analysis:** Understanding market trends and predicting housing bubbles.

Dataset Description

For this example, we use a simple dataset with the following features:

- **Square Footage:** The size of the house in square feet.
- **Number of Bedrooms:** The number of bedrooms in the house.
- **Location Score:** A score representing the desirability of the house's location.
- **Price:** The target variable representing the price of the house.

Implementing the Model in Python

Step 1: Importing Required Libraries

```
import pandas as pd
from sklearn.ensemble import RandomForestRegressor from sklearn.model_selection import
train_test_split from sklearn.metrics import mean_absolute_error
```

Step 2: Preparing the Dataset

We create a simple dataset representing house features and their corresponding prices: # Example dataset

```
data = {
    'square_feet': [1500, 2000, 2500, 3000, 3500],
    'bedrooms': [3, 4, 4, 5, 5],
    'location_score': [7, 8, 8, 9, 10],
    'price': [300000, 400000, 500000, 600000, 700000]

                                                    }

df = pd.DataFrame(data)
X = df[['square_feet', 'bedrooms', 'location_score']]
y = df['price']
```

Step 3: Splitting the Data

Splitting the dataset into training and testing subsets ensures the model can be evaluated on unseen data: # Split data

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Step 4: Training the Model